Research Report

Artificial intelligence in healthcare refers to the application of artificial intelligence (AI) to analyze and understand complex medical and healthcare data. It can often augment and in some cases exceed human capabilities by providing better or faster ways to diagnose, treat, or prevent disease. Research is ongoing into its applications across various medical subdisciplines and related industries.

resents unprecedented ethical concerns related to issues such as data privacy, automation of jobs, and amplifying already existing algorithmic bias. New technologies such as AI are often met with resistance by healthcare leaders, leading to slow and erratic adoption. Research has identified inadequate implementation infrastructure, limited workflow integration, and the absence of ongoing performance monitoring as primary barriers.

AI can assist clinicians with data processing capabilities to save time and improve accuracy. Machine learning can substantially aid doctors in patient diagnosis through the analysis of mass electronic health records. AI can help early prediction, for example, of Alzheimer's disease and dementias, by looking through large numbers of similar cases and possible treatments.

Artificial intelligence (AI) is everywhere these days, including healthcare. No chatbot or large language model (LLM) is a substitute for expert medical care. Instead, AI isgiving healthcare teams better tools to care for you, and it can make a big difference for your well-being.

AI is being used in healthcare to allow people to work more effectively and efficiently. "AI is no longer an experiment," says Ben Shahshahani, PhD, Cleveland Clinic's Chief AI Officer. Here are some of the ways it may help your providers help